

Project Title

Analyzing the Impact of Content Features on User Engagement on Instagram

Motivation

With the increasing role of social media in marketing and communication, understanding what drives user engagement has become essential. Platforms such as Instagram provide measurable engagement metrics such as likes and comments, allowing data-driven analysis of user behavior. Engagement metrics are widely used as indicators of content performance and audience interest.

Previous research has shown that content characteristics significantly influence engagement. For example, factors such as content type and interactivity affect the number of likes and comments on posts (de Vries et al., 2012). These findings suggest that engagement is not random but influenced by observable features.

Therefore, this project aims to analyze how different content-related features affect user engagement on Instagram and provide insights for improving content strategies.

Research Question

To what extent can content-related features explain and predict user engagement on Instagram?

Data Sources

1. Instagram Influencers Dataset

- Source: Kaggle – Global Instagram Influencers Ranking Dataset
<https://www.kaggle.com/datasets/wardabilal/global-instagram-influencers-ranking-dataset>
- Description:
This dataset consists of approximately 200 Instagram influencers and 10 variables, including follower count, number of posts, average likes, engagement rate, and country.
- Data Collection:
The dataset will be downloaded from Kaggle and imported into Python for analysis. Data preprocessing will include handling missing values and ensuring consistency across variables to improve data quality.

2. Academic Source

- Source: de Vries et al. (2012)
<https://doi.org/10.1016/j.intmar.2012.01.003>
- Description:
This study provides a theoretical foundation for analyzing how content characteristics influence engagement on social media. It shows that features such as content type and interactivity significantly affect engagement metrics such as likes and comments, supporting the approach used in this project.

Feature Engineering & Data Enrichment

To enrich the dataset, additional features will be created from existing variables. These include:

- Total engagement (likes + comments)
- Engagement per post
- Engagement normalized by follower count
- Posting activity measures (based on total posts)

These derived variables enhance the dataset by capturing deeper interaction patterns that are not directly observable from raw features. This enrichment step is important for improving the explanatory and predictive power of the analysis.

Methodology

The project will follow a standard data science pipeline. First, the dataset will be preprocessed by handling missing values and preparing variables for analysis to ensure data quality and reliability.

Next, exploratory data analysis (EDA) will be conducted to examine relationships between variables such as follower count, engagement, and content category. Correlation analysis will also be applied to identify the most relevant features.

Finally, machine learning models will be used to analyze engagement. A regression model will predict engagement levels, while a classification model will categorize posts into high and low engagement groups.

Expected Outcomes

The project is expected to identify key factors that influence user engagement on Instagram. It will also demonstrate how accurately engagement can be predicted using content-related features and provide insights that can support more effective content strategies.